

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1012

Duck Creek 138 kV
March 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1012. The Interconnection Customer proposes a 826.4 MW natural gas combined cycle facility interconnecting at the Duck Creek 138 kV substation in the Pennsylvania area. The request was received on June 06, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 314 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$162,855,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cico / Duck Creek Substation 138 kV L	Duck Crk / Bayfield Substation 138 kV Line	282.9	267	63.2
Piney / Bayfield 138 kV	Cico / Duck Creek 138 kV L	264.9	339	75.5
Piney / Piney Creek Project 138 kV	Cico / Duck Creek 138 kV	263.1	291	65.5
Duck Crk / Bayfield 138 kV	Cico / Duck Creek 138 kV	259.4	349	75.5
Malvern / Macon Solar Power Project 138 kV	Duck Creek / Bayfield 138 kV	255.2	187	36.8
Bloomfield / Malvern 138 kV Line	Bayfield / Duck Creek 138 kV	244.4	192	36.8
Duck Creek / Macon Solar Power Project 138 kV	Duck Crk / Bayfield Substation 138 kV	244.3	197	36.8
Larene / Piney Crk Project 138 kV line	Caldwell / Piney Creek Project 345 kV	209.4	223	11.2
G W Ivey / Divide Substation 138 kV L	Piney Crk Project / Piney 138 kV L	181.4	180	28.0
Piney / G W Ivey 138 kV	Piney / Piney Creek Project 138 kV	180.2	185	28.0
Raun / Divide 138 kV	Piney Crk Project / Piney 138 kV	163.4	198	28.0
Piney Crk Project / Geysers Tap 1 138 kV	Piney Crk Project / Piney 138 kV line	161.8	222	31.2
Piney Crk Project 345 kV/138 kV Transformer	Piney Crk Project 500/345kV Autotransformer	154.2	353	36.6
Raun / Geysers Tap 1 138 kV	Piney / Piney Crk Project 138 kV	150.7	212	28.0
Bloomfield / Glen Avon 138 kV	Caldwell / Piney Crk Project 345 kV Line	144.2	225	10.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hughtron / Stockton Wyndham 138 kV Line	Caldwell / Piney Crk Project 345 kV Line	141.9	235	6.8
Permanente / Hughtron Substation 138 kV	Caldwell / Piney Crk Project 345 kV line	140.5	240	6.8
Hillabee Station / Glen Avon 138 kV L	Caldwell / Piney Crk Project 345 kV	139.6	230	10.1
Bulls Gap / S Springfield 138 kV	Allegheny Hydro No 8 345/138 kV XFMR	138.2	228	10.0
Larene / Permanente 138 kV Line	Piney Crk Project / Caldwell 345 kV	138.1	232	6.7
Bulls Gap / E B Harris Electric Generating Plant 138 kV line	Allegheny Hydro No 8 345/138kV Transformer Bank	135.2	231	10.0
Allegheny Hydro No 9 / Allegheny Hydro No 8 138 kV	Allegheny Hydro No 8 / Corydon 138 kV Line	134.6	252	5.1
E B Harris Electric Generatingplant / Big Cajun 2 Substation 138 kV	Allegheny Hydro No 8 345/138 kV Auto	131.4	234	10.0
Big Cajun 2 / Southeast Vicksburg 138 kV L	Allegheny Hydro No 8 345/138 kV XFMR	123.4	243	10.0
Allegheny Hydro No 8 / Corydon 138 kV	Piney Crk Project / Caldwell 345 kV L	116.1	254	14.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Duck Creek / Cico 138 kV Line	Rebuild line with higher-capacity conductor	\$26,145,000	67% (\$17,423,000)
Bayfield / Piney 138 kV	Rebuild line with higher-capacity conductor	\$3,894,000	67% (\$2,595,000)
Piney / Piney Crk Project 138 kV line	Rebuild line with higher-capacity conductor	\$3,312,000	67% (\$2,207,000)
Duck Crk / Bayfield 138 kV	Rebuild line with higher-capacity conductor	\$3,013,000	67% (\$2,008,000)
Malvern / Macon Solar Power Project 138 kV line	Rebuild line with higher-capacity conductor	\$4,716,000	67% (\$3,143,000)
Bloomfield / Malvern 138 kV Line	Rebuild line with higher-capacity conductor	\$33,010,000	67% (\$21,997,000)
Duck Creek / Macon Solar Power Project 138 kV	Rebuild line with higher-capacity conductor	\$7,272,000	67% (\$4,846,000)
Larene / Piney Crk Project 138 kV	Rebuild line with higher-capacity conductor	\$4,307,000	67% (\$2,867,000)
G W Ivey / Divide Substation 138 kV L	Rebuild line with higher-capacity conductor	\$17,838,000	67% (\$11,888,000)
G W Ivey / Piney 138 kV	Rebuild line with higher-capacity conductor	\$18,541,000	67% (\$12,356,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Divide / Raun 138 kV	Rebuild line with higher-capacity conductor	\$11,177,000	67% (\$7,448,000)
Piney Creek Project Substation / Geysers Tap 1 138 kV Line	Rebuild line with higher-capacity conductor	\$3,841,000	67% (\$2,560,000)
Piney Crk Project 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$23,283,000	60% (\$14,070,000)
Raun / Geysers Tap 1 138 kV	Rebuild line with higher-capacity conductor	\$3,724,000	67% (\$2,481,000)
Bloomfield / Glen Avon 138 kV	Rebuild line with higher-capacity conductor	\$15,565,000	51% (\$7,993,000)
Hughtron / Stockton Wyndham 138 kV Line	Rebuild line with higher-capacity conductor	\$25,201,000	66% (\$16,770,000)
Permanente / Hughtron Substation 138 kV	Rebuild line with higher-capacity conductor	\$8,790,000	66% (\$5,850,000)
Glenavon / Hillabe Station 138 kV	Rebuild line with higher-capacity conductor	\$5,903,000	51% (\$3,031,000)
Bulls Gap / S Springfield 138 kV	Rebuild line with higher-capacity conductor	\$3,790,000	67% (\$2,523,000)
Larene / Permanente 138 kV	Rebuild line with higher-capacity conductor	\$5,280,000	66% (\$3,513,000)
E B Harris Electric Generating Plant / Bulls Gap 138 kV	Rebuild line with higher-capacity conductor	\$1,530,000	67% (\$1,018,000)
Allegheny Hydro No 9 / Allegheny Hydro No 8 138 kV	Rebuild line with higher-capacity conductor	\$8,194,000	66% (\$5,445,000)
E B Harris Electric Generatingplant / Big Cajun 2 Substation 138 kV	Rebuild line with higher-capacity conductor	\$6,320,000	67% (\$4,207,000)
Southeast Vicksburg / Big Cajun 2 138 kV	Rebuild line with higher-capacity conductor	\$3,182,000	67% (\$2,119,000)
Allegheny Hydro No 8 / Corydon 138 kV	Rebuild line with higher-capacity conductor	\$3,749,000	67% (\$2,497,000)
	Total allocated to project		\$162,855,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.